MCGI: Manifold-Consistent Graph Indexing for Billion-Scale Disk-Resident Vector Search

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Abstract

Graph-based Approximate Nearest Neighbor (ANN) search often suffers from performance degradation in high-dimensional spaces due to the Euclidean-Geodesic mismatch, where greedy routing diverges from the underlying data manifold. To address this challenge, this paper presents Manifold-Consistent Graph Indexing (MCGI), a geometry-aware and disk-resident indexing method that leverages Local Intrinsic Dimensionality (LID) to dynamically adapt search strategies to the intrinsic geometry of data. Unlike conventional algorithms that treat dimensions uniformly, MCGI modulates its beam search budget based on in-situ geometric analysis, which reduces sensitivity to data-specific hyperparameters by replacing a single scalar with a geometry-informed range that remains stable across datasets of varying dimensionality. Theoretical analysis demonstrates that MCGI provides robust approximation by preserving manifold-consistent topological connectivity. Extensive evaluations against five industry-standard baselines across five datasets up to billion scales confirm the advantages of the proposed approach.

1 Introduction

The advent of Large Language Models [4, 39] has fundamentally transformed the landscape of information retrieval and knowledge management. To address the inherent limitations of LLMs, such as hallucinations [19] and knowledge cutoff dates, Retrieval-Augmented Generation (RAG) [27] has emerged as a critical architectural paradigm. RAG relies heavily on the ability to retrieve semantically relevant context from massive corpora in real-time, typically via dense vector representations [24]. This dependency has placed Approximate Nearest Neighbor Search (ANNS) at the core of modern data infrastructure, demanding vector indices that can scale to billion-point datasets [35] while maintaining low latency and high recall under strict production constraints.

For extreme-scale embedding vector applications, many ANNS techniques have noticeably converged on graph-based indices, with DiskANN (Vamana) and its variants [36, 5, 11, 12] being representative solutions for large-scale, SSD-resident workloads. These algorithms typically employ greedy routing on a proximity graph to navigate from an entry point to the query target. While such methods exhibit exceptional performance on popular benchmarks like SIFT1M [21] (128 dimensions), their efficiency degrades significantly in high-dimensional spaces, such as GIST1M (960 dimensions). This degradation is often attributed to the curse of dimensionality [2], where the distance contrast diminishes, and the Euclidean shortest path on the graph diverges from the geodesic path on the underlying data manifold [37, 18]. We refer to this phenomenon as the *Euclidean-Geodesic Mismatch*, or EGM. When the routing algorithm ignores the intrinsic geometry of the data, it performs excessive backtracking and disk I/O, rendering the search inefficient, which is particularly problematic in high-recall scenarios where production systems often expect Recall@10 higher than 95%.

Our observation on this EGM problem is that high-dimensional real-world data is rarely uniformly distributed; Instead, it typically adheres to the Manifold Hypothesis [37, 33], residing on lower-

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dimensional structures embedded within the ambient space. Consequently, the search difficulty is not uniform across the dataset but is modulated by the Local Intrinsic Dimensionality (LID) [26]. That is, in regions where the data manifold is flat (low LID), greedy routing is effective; however, in regions with high curvature or complex topology (high LID), standard greedy strategies may fail to identify the descent direction. *We thus argue that an optimal indexing strategy must be manifold-aware, dynamically allocating computational resources based on the local geometric complexity.*

To demonstrate the above hypothesis, we design Manifold-Consistent Graph Indexing (MCGI), a geometry-aware disk-based indexing method crafted to align Euclidean search with the underlying data manifold. By incorporating LID estimation directly into the routing logic, MCGI demonstrates the ability to adapt its traversal strategy to the local topology of the underlying manifold.

The remainder of this paper will present the following contributions: (1) A theoretical framework embedding the local intrinsic dimensionality to graph navigability with formal guarantees on estimation robustness, fixed-beam suboptimality, and construction-time adaptation; (2) Geometry-aware indexing algorithms that adapt pruning parameters without data-specific hyperparameters; and (3) Up to $5.8\times$ higher throughput on high-dimensional vectors (GIST1M, 960 dimensions) and $3\times$ lower latency on billion-scale vectors (e.g., SIFT1B) against five baseline indexing methods.

2 Related Work

Vector Indexing Paradigms While traditional sparse retrieval methods rely on lexical matching [32], memory-resident graph indices like HNSW [29] establish state-of-the-art dense retrieval but face billion-scale memory bottlenecks. Disk-based approaches, such as DiskANN [36], SPANN [5], PipeANN [11], and OdinANN [12], along with predecessors like NSG [9], adapt topologies or inverted structures for SSDs. Recent advancements such as CSPG [41] further optimize sparse proximity graphs to reduce unpromising explorations, yet these methods predominantly rely on static routing parameters evaluated on moderate dimensionality. In contrast, MCGI dynamically modulates the search with local topology to align graph traversal with the underlying manifold.

High-Dimensional Indexing Early space-partitioning trees like the KD-tree [3] and R-tree [14] severely degrade to linear-scan performance in dimensions higher than 20 due to the curse of dimensionality. Approximate methods address this through sub-linear hashing like LSH [17, 7], which incurs excessive storage overhead, or subspace quantization like PQ [21] and OPQ [10], and randomized structures like RP-Trees [6], which sacrifice accuracy through compression. Furthermore, recent literature extensively studies the theoretical limits of navigable graphs [8] and the efficiency of index construction [43] in strictly high-dimensional scenarios. MCGI circumvents aggressive quantization losses and redundant hashing overheads by preserving the exact connectivity benefits of graph traversal adapted to local topology.

Intrinsic Dimensionality Theoretical metrics such as the doubling dimension [13] and expansion dimension [23] provide asymptotic bounds on search complexity, and recent analyses establish worst-case performance limits for popular implementations like DiskANN and HNSW based on intrinsic dimensionality [18]. Maximum likelihood estimators [26] for Local Intrinsic Dimensionality (LID) enable robust practical estimations. Existing works typically utilize LID passively for query hardness prediction [15, 40] or detecting adversarial examples [28] without altering the underlying index structure. Diverging from passive descriptive applications, MCGI actively utilizes LID as a prescriptive control signal to dynamically modulate graph traversal parameters for efficient routing.

3 Methodology

3.1 Mapping Function from LID to Pruning Parameter

We start by briefly reviewing the notions of Local Intrinsic Dimensionality (LID) in the words of analysis. A full treatment can be found, for example, in Houle [16].

Definition 3.1 (Local Intrinsic Dimensionality)**.** Let $\mathcal{X}$ be a domain equipped with a distance measure $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}^+$. For a reference point $x \in \mathcal{X}$, let $F_x(r) = \mathbb{P}(d(x, Y) \leq r)$ denote the cumulative distribution function (CDF) of the distance between x and a random variable Y drawn from the

underlying data distribution, where $\mathbb{P}$ denotes the probability function and r denotes the radius. The Local Intrinsic Dimensionality (LID) of x, denoted as $\mathrm{LID}(x)$, is defined as the intrinsic growth rate of the probability measure within the neighborhood of x:

$$\mathrm{LID}(x) \triangleq \lim_{r \to 0} \frac{r \cdot F'_x(r)}{F_x(r)} = \lim_{r \to 0} \frac{d \ln F_x(r)}{d \ln r}, \tag{1}$$

provided that the limit exists and $F_x(r)$ is continuously differentiable for $r > 0$.

While Definition 3.1 is elegant, it is often impractical to compute directly. According to [1], LID can be estimated as follows.

Definition 3.2 (LID Maximum Likelihood Estimator)**.** Given a reference point x and its k-nearest neighbors determined by the distance measure d. Let $r_i = d(x, v_i)$ denote the distance to the i-th nearest neighbor, sorted such that $r_1 \leq \cdots \leq r_k$. Following the formulation in [1], which adapts the Hill estimator for intrinsic dimensionality, the LID at x can be estimated as:

$$\widehat{\mathrm{LID}}(x) = -\left(\frac{1}{k} \sum_{i=1}^{k} \ln \frac{r_i}{r_k} \right)^{-1}. \tag{2}$$

The primary hypothesis of the proposed Manifold-Consistent Graph Indexing (MCGI) is that the graph topology should adapt to the local geometric complexity, rather than being a static vertex-edge structure. To preserve topological fidelity, the indexing algorithm must adopt a conservative pruning strategy (smaller α in [36]), thereby forcing the search to take smaller, safer steps along the manifold surface. The question then becomes how to quantitatively map the estimated LID to the pruning parameter α that controls the edge selection during graph construction. We refer to such a function as the *mapping function* $\Phi : \mathbb{R}^+ \to [\alpha_{\min}, \alpha_{\max}]$, which translates the geometric complexity of the local neighborhood into a corresponding pruning constraint.

Formally, let $u \in V$ be a node in the graph and $\widehat{\mathrm{LID}}(u)$ be its estimated LID. We define the *pruning parameter* $\alpha(u)$ as:

$$\alpha(u) \triangleq \Phi(\widehat{\mathrm{LID}}(u)), \tag{3}$$

where Φ will be constructed to satisfy the following geometric requirements: in regions with high LID, the graph should enforce a stricter connectivity constraint (smaller α) to avoid short-circuiting the manifold; conversely, in low-LID regions, the constraint can be relaxed (larger α).

To ensure the mapping is robust across datasets with varying complexity scales, we employ Z-score normalization based on the empirical distribution of the LID estimates. We first compute the normalized score $z(u)$:

$$z(u) = \frac{\widehat{\mathrm{LID}}(u) - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}, \tag{4}$$

where $\mu_{\widehat{\mathrm{LID}}}$ and $\sigma_{\widehat{\mathrm{LID}}}$ denote the mean and standard deviation of the set of estimated LID values $\{\widehat{\mathrm{LID}}(v) \mid v \in V\}$ computed across the entire graph. We then formulate Φ using a logistic function to smoothly map the Z-score to the operational range $[\alpha_{\min}, \alpha_{\max}]$:

$$\Phi(\widehat{\mathrm{LID}}(u)) = \alpha_{\min} + \frac{\alpha_{\max} - \alpha_{\min}}{1 + \exp(z(u))}. \tag{5}$$

The mapping function Φ satisfies the following geometric properties that are essential for stable graph construction: Monotonicity and Boundedness. We state these properties in the following as propositions with proof sketches and provide full proofs in the Appendix.

Proposition 3.3 (Monotonicity)**.** *The mapping function Φ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{6}$$

Proof Sketch. By the chain rule, $\frac{d\Phi}{d\widehat{\text{LID}}(u)} = \frac{d\Phi}{dz} \cdot \frac{dz}{d\widehat{\text{LID}}(u)}$, where the Z-score satisfies $\frac{dz}{d\widehat{\text{LID}}(u)} = \frac{1}{\sigma_{\widehat{\text{LID}}}} > 0$. The derivative of the logistic component evaluates to $\frac{d\Phi}{dz} = -(\alpha_{\max} - \alpha_{\min}) \cdot \frac{e^z}{(1+e^z)^2}$, which is strictly negative since $\alpha_{\max} > \alpha_{\min}$ and $e^z > 0$ for all $z \in \mathbb{R}$. Combining these two terms yields a strictly negative derivative: $\alpha(u)$ decreases monotonically as geometric complexity increases. □

Proposition 3.4 (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. That is, for any node u with a finite LID estimate, the following holds:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{7}$$

Proof Sketch. For any finite $\widehat{\text{LID}}(u)$, the Z-score $z(u)$ is finite, and the logistic function satisfies $S(u) = \frac{1}{1+e^{z(u)}} \in (0, 1)$ strictly, since $e^{z(u)} \in (0, \infty)$. Substituting into $\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u)$ and applying the strict bounds on $S(u)$ directly yields $\alpha_{\min} < \alpha(u) < \alpha_{\max}$. □

3.2 Indexing Algorithms

The Manifold-Consistent Graph Indexing (MCGI) algorithm introduces a geometric calibration phase to a graph indexing procedure. Unlike static methods that apply a uniform connectivity rule, MCGI executes in two distinct stages to ensure that the topology respects the manifold structure, as follows. The full pseudo-code of the algorithm is in Appendix A.3, Algorithm 1.

Phase 1: Geometric Calibration. Before modifying the graph topology, the system first performs a global analysis of the dataset geometry. We estimate the LID for every point and aggregate the population statistics (μ, σ) defined in Section 3.1. This phase freezes the geometric profile of the dataset. By pre-computing these statistics, we decouple the complexity estimation from the graph update loop, ensuring that the mapping function Φ remains stable and computationally efficient during the intensive edge-selection process.

Phase 2: Manifold-Consistent Refinement. The index construction follows an iterative refinement strategy. Let $\mathcal{N}(u)$ denote the set of neighbors for node u in the graph G. In each iteration, the algorithm dynamically updates $\mathcal{N}(u)$ by considering a candidate pool $\mathcal{C}$ obtained from a greedy search on the current graph structure:

1. Queries the pre-computed geometric profile to determine the node-specific constraint $\alpha(u)$.
2. Explores the graph to identify a candidate pool $\mathcal{C}$.
3. Filters connections using the dynamic occlusion criterion.

Remark The calibration phase takes $O(N \log N)$ dominated by the construction phase, on par with the complexity of DiskANN [36]. In practice, calculating the exact LID for the entire dataset may cause a computational bottleneck on billion-scale data. To address this, we also design an online algorithm called *Online-MCGI*, whose pseudo-code can be found in Appendix A.3, Algorithm 2.

3.3 Theoretical Analysis

3.3.1 Geometric Complexity and Adaptive Routing

The efficiency of greedy routing on a proximity graph is intrinsically tied to the local topological properties of the underlying manifold $\mathcal{M}$. Under the Manifold Hypothesis, the search space is locally characterized by the Local Intrinsic Dimensionality (LID) at query q. We formalize the relationship between LID and the computational cost required to identify the nearest neighbor, as follows.

Lemma 3.5 (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query q on a manifold $\mathcal{M}$, the expected number of distance evaluations N_{dist} required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = LID(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{8}$$

where $\lambda > 0$ is a geometric constant derived from the required convergence rate.

Proof Sketch. At a node u with distance r to query q, a greedy step succeeds only if a neighbor v falls within a cone of half-angle $\theta_{\max}$ around the direction of q; the probability of this event corresponds to the ratio of a spherical cap to the full hypersphere $\mathcal{S}^{d-1}$. Asymptotic analysis of this ratio via the regularized incomplete beta function yields $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$, where the exponential decay arises from the concentration of measure on high-dimensional spheres and $\lambda = -\ln(\sin\theta_{\max}) > 0$. Since $\mathbb{E}[N_{dist}] = 1/P_{\text{success}}$, the lower bound follows directly. □

Lemma 3.5 reveals a fundamental geometric barrier: for a fixed budget L, the probability of routing failure grows exponentially in high-LID regions. To address this, MCGI adopts an *Iso-Recall* strategy, aiming to homogenize the search reliability across the heterogeneous manifold.

Proposition 3.6 (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability δ across the manifold (i.e., $\mathbb{P}(\textit{fail} \mid q) \le \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda \cdot \textit{LID}(q)\right). \tag{9}$$

Proof Sketch. For a beam of width L, the failure probability is $(1 - P_{\text{success}})^L \le \delta$, which via the approximation $\ln(1-x) \approx -x$ requires $L \ge -\ln\delta / P_{\text{success}}$. Substituting the bound $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$ from Lemma 3.5 and absorbing the polynomial and confidence factors into a constant C_δ yields $L(q) \ge C_\delta \cdot \exp(\lambda \cdot \text{LID}(q))$. □

While Proposition 3.6 establishes that the search budget L must theoretically adapt to local geometric complexity, dynamically modulating L per node during online serving disrupts memory alignment and degrades SIMD parallel efficiency. Therefore, in practical system deployments, L is strictly bounded as a constant.

3.3.2 Topological Fidelity and Connectivity

A primary theoretical concern with aggressive edge pruning is the potential fracture of the connectivity backbone [42]. We will show that MCGI guarantees global reachability by strictly preserving the underlying manifold skeleton.

Formally, the edge selection in MCGI is governed by the adaptive pruning parameter $\alpha(u)$. Specifically, an edge (u, v) is pruned if there exists a witness node n such that $\alpha(u) \cdot d(n, v) \le d(u, v)$. This condition defines an *exclusion region* around the midpoint of u and v. As $\alpha(u)$ increases, this region shrinks, making pruning more conservative. We formulate this in the following proposition.

Proposition 3.7 (Connectivity Preservation)**.** *Let G_{EMST} and G_{RNG} denote the Euclidean Minimum Spanning Tree (EMST) and the Relative Neighborhood Graph (RNG), respectively. For any configuration of points in general position, provided that $\alpha(u) \ge 1.0$ for all $u \in V$, the graph G_{MCGI} satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{10}$$

Consequently, G_{MCGI} is connected.

Proof Sketch. The first inclusion $E_{EMST} \subseteq E_{RNG}$ follows from the classical result of Toussaint [38], which holds for any finite point set in a metric space. For the second inclusion $E_{RNG} \subseteq E_{MCGI}$, note that MCGI prunes an edge (u, v) only if a witness node falls within a exclusion region of radius $\frac{1}{\alpha(u)}d(u, v)$ around v; since $\alpha(u) \ge 1.0$, this exclusion region is a strict geometric subset of the RNG exclusion region, so any edge surviving the RNG pruning condition is guaranteed to survive MCGI pruning. Connectivity of G_{MCGI} follows immediately from the inclusion hierarchy, since G_{EMST} spans all vertices by definition. □

Proposition 3.7 ensures that there are no structural dead ends. While Lemma 3.5 dictates the *cost* of routing, Proposition 3.7 ensures the *viability* of routing. Even in the worst-case scenario where heuristics fail, a path exists from the entry point to any target via the edges of the underlying EMST, provided the graph traversal algorithm has sufficient width to discover these backbone links.

3.3.3 Robustness to LID Estimation Error

A practical concern with the LID-based mapping is whether estimation noise from the finite-sample MLE (Definition 3.2) propagates into the pruning parameter $\alpha(u)$ in a destabilizing manner. We address this concern in the following.

The MLE estimator in Definition 3.2 is known to exhibit asymptotic variance of order LID^2/k [26], yielding a pointwise estimation error bounded as:

$$\left|\widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)\right| = O\left(\frac{\mathrm{LID}(u)}{\sqrt{k}}\right). \tag{11}$$

We characterize how this error propagates through the mapping function Φ.

Proposition 3.8 (Estimation Robustness)**.** *Let $\widehat{\mathrm{LID}}(u)$ be the MLE estimate computed with k nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{12}$$

Proof Sketch. By a first-order Taylor expansion, $|\Delta\alpha(u)|$ is bounded by $\sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|$, where $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ is the estimation error. The gradient magnitude is bounded by $\frac{\alpha_{\max}-\alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}$, since the logistic derivative $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $\frac{1}{4}$ at $z = 0$, providing an intrinsic noise attenuation factor regardless of LID magnitude. Substituting the known asymptotic MLE error $|\epsilon| = O\left(\mathrm{LID}(u)/\sqrt{k}\right)$ [26] yields the bound directly, confirming that $|\Delta\alpha(u)|$ decays as $O(1/\sqrt{k})$ and even modest k suffices for stable α assignments. □

Proposition 3.8 reveals two structural properties. First, the perturbation decays as $O(1/\sqrt{k})$, confirming that even a modest neighborhood size (e.g., $k = 20$) provides stable α assignments. Second, the Sigmoid mapping contributes a factor of at most $1/4$ (the global maximum of the logistic derivative), acting as an intrinsic noise buffer regardless of the LID magnitude. This provides a principled lower bound on the required k during the geometric calibration phase, and explains the empirical stability that we will discuss in §4.5 across varying dataset geometries.

3.3.4 Theoretical Justification of Construction-time Adaptation

While Proposition 3.6 establishes that an *ideal* search system should allocate beam width $L^*(q) \propto \exp(\lambda \cdot \mathrm{LID}(q))$ at query time, implementing this schedule online is impractical due to SIMD alignment and memory access constraints (§3.3.1). We now formalize the cost of abandoning this schedule and show that how MCGI's construction-time adaptation recovers the theoretical optimum.

Definition 3.9 (Iso-Recall Beam Schedule)**.** For a target failure probability δ, the oracle beam schedule is defined as:

$$L^*(q) = C_\delta \cdot \exp(\lambda \cdot \mathrm{LID}(q)), \tag{13}$$

where $C_\delta = -\ln\delta / P_{\text{success}}$ is a recall-dependent constant. A search algorithm is said to be *recall-optimal* if it implements $L^*(q)$ for every query q.

Proposition 3.10 (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies:*

$$P(\text{fail} \mid q) - \delta \;\geq\; 1 - \exp\left(-\frac{C_\delta}{\bar{L}}\exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\text{avg}}\big)\big)\right), \tag{14}$$

which grows exponentially with the deviation of $\mathrm{LID}(q)$ from the dataset mean $\mathrm{LID}_{\text{avg}}$.

Proof Sketch. For a fixed beam width $\bar{L}$, the failure probability is $(1 - P_{\text{success}})^{\bar{L}}$, while the oracle schedule $L^*(q) = C_\delta \exp(\lambda \cdot \mathrm{LID}(q))$ (Definition 3.9) achieves exactly δ by construction; the recall gap is thus the difference between these two quantities. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\text{avg}}$, the shortfall $\Delta L(q) = L^*(q) - \bar{L} > 0$ grows exponentially with $\mathrm{LID}(q)$, and applying the inequality $1 - e^{-x} \leq x$ to the resulting expression yields the closed-form lower bound in Eq. (14). The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, so the recall deficit grows exponentially, creating a widening gap that cannot be compensated by any fixed-parameter strategy. □

Proposition 3.10 quantifies precisely why static-parameter methods degrade on high-dimensional data: the recall deficit is not a constant offset but an *exponentially widening gap*. This provides a theoretical justification for geometry-aware adaptation that goes beyond empirical motivation. In practice, the following corollary establishes that MCGI's construction-time adaptation monotonically reduces this gap compared to any fixed-parameter baseline, structurally compensating for the shortfall of a constant beam width.

Corollary 3.11 (Construction-time Structural Compensation for Fixed Beam Width)**.** *Under any fixed beam width $\bar{L}$, modulating $\alpha(u)$ during index construction according to $\Phi(\widehat{\mathrm{LID}}(u))$ monotonically reduces $P(\mathrm{fail} \mid q)$ for all queries in high-LID regions compared to any static-α baseline operating under the same $\bar{L}$. Specifically, nodes in high-LID regions receive conservative pruning (small α), preserving a strictly larger navigational edge set that structurally compensates for the shortfall of $\bar{L}$ relative to the oracle schedule $L^*(q)$. This guarantee is monotone but not tight: it establishes that MCGI dominates fixed-α baselines, rather than quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) directly.*

Proof Sketch. For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\mathrm{avg}}$, Proposition 3.3 guarantees $\alpha(u) < \bar{\alpha}$, which strictly shrinks the pruning exclusion region and thus preserves a superset of edges relative to any static-$\bar{\alpha}$ baseline, i.e., $E_{\mathrm{MCGI}} \supseteq E_{\bar{\alpha}}$ in high-LID regions. A strictly larger outgoing edge set monotonically increases $P_{\mathrm{success}}(q)$ at each beam step by the same geometric argument as Proposition 3.6, yielding $P(\mathrm{fail} \mid q, \mathrm{MCGI}) \leq P(\mathrm{fail} \mid q, \bar{\alpha})$ for all q with $\mathrm{LID}(q) > \mathrm{LID}_{\mathrm{avg}}$. Note that this guarantee is monotone but not tight: quantifying the residual gap to the oracle schedule $L^*(q)$ (Definition 3.9) is left for future work. □

4 Evaluation

We implemented MCGI with C++; some of the code has been merged into DiskANN [36]. The anonymized code is at: `https://anonymous.4open.science/r/MCGI/`. Reported numbers are averaged over 3 independent runs with unnoticeable variance, thereby not displayed in the figures.

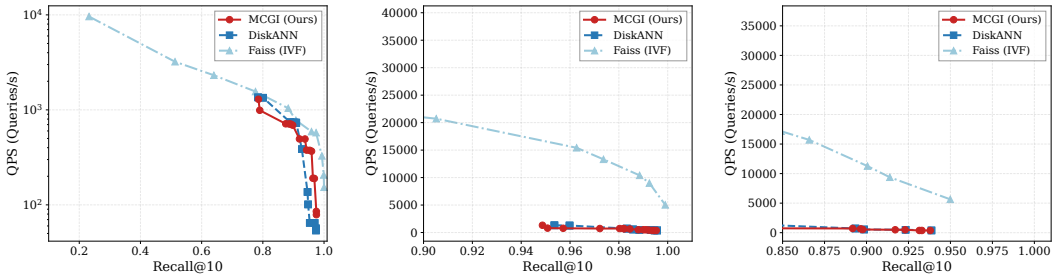
All experiments were conducted with 96 CPU cores, 256 GiB of RAM, and 7.68 TB of NVMe SSD on Chameleon Cloud [25]. We evaluate MCGI on five benchmarks ranging from million-scale to billion-scale: SIFT1M [21], GloVe-100 [30], GIST1M [21], SIFT1B [22], and T2I-1B [34]. Our evaluation compares MCGI against five baselines: PipeANN [12, 11], CSPG [41], DiskANN (Vamana) [36], IVF-Flat (Faiss) [20], and HNSW [29]. More setup details are in Appendix A.2.

4.1 High-dimensional Effectiveness

While the in-memory Faiss baseline establishes a theoretical throughput upper-bound, disk-resident indices face severe constraints from SSD random I/O latency. As illustrated in Figure 1a, MCGI mitigates this bottleneck on the high-dimensional GIST1M dataset by leveraging manifold-aware routing to minimize necessary disk reads, achieving 375 QPS at 95% recall and delivering a 5.8× speedup over the baseline DiskANN. Furthermore, evaluations on lower-dimensional datasets like SIFT1M and GloVe-100 (Figures 1b and 1c) confirm that MCGI achieves performance parity with DiskANN. These results indicate that our adaptive mechanism robustly reduces to standard greedy search on simpler geometries without incurring any computational overhead, narrowing the gap between disk-based scalability and in-memory efficiency.

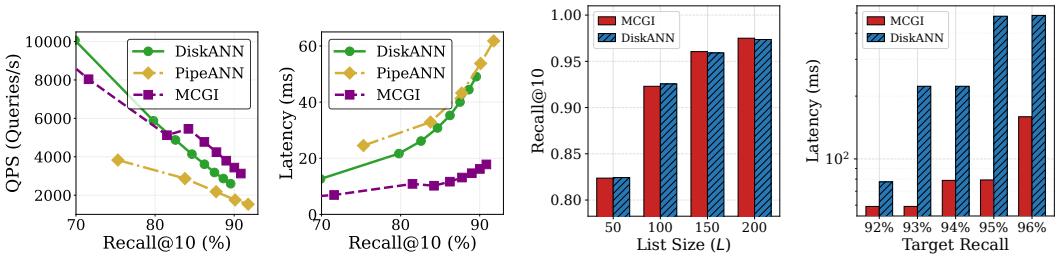
4.2 High-recall Efficiency

Table 1 reports peak QPS of different indexing methods on high-dimensional vectors (i.e., 960-dimensional GIST1M) at Recall@10 $\geq$ 95%, where R denotes the maximum out-degree of each graph node. Disk-resident baselines plateau well below 100 QPS: DiskANN reaches only 64.7 QPS, and PipeANN's aggressive I/O parallelism yields at most 60.3 QPS at $R = 48$, confirming that system-level optimizations cannot compensate for topologically suboptimal routing in high-dimensional spaces. MCGI delivers 375.1 QPS, a 5.8× gain over DiskANN and 6.2× over PipeANN-R48. Notably, MCGI reaches 64% of the throughput of the memory-resident IVF Flat index, narrowing the historical performance gap between disk-resident and in-memory indices without sacrificing the storage efficiency of SSD-based deployment.



(a) GIST1M (960-dim) (b) SIFT1M (128-dim) (c) GloVe (100-dim)

Figure 1: Recall-QPS trade-off on popular datasets of various dimensions.



(a) Performance trade-off (SIFT1B, Online-MCGI) (b) Sensitivity (GIST1M) (c) Latency (GIST1M)

Figure 2: Scalability and Hyperparameter Sensitivity.

4.3 Scalability and Hyperparameter Sensitivity

We restrict billion-scale comparisons to PipeANN and DiskANN, which share the same Vamana backbone as MCGI and thus isolate the contribution of manifold-aware routing. All reported results on billion-scale datasets use Online-MCGI (Algorithm 2), where LID estimation is performed on-the-fly using the candidate set discovered during greedy search. The geometric calibration overhead is therefore fully included in the reported latency numbers. As shown in Figure 2a, at 90% recall MCGI maintains 3,436 QPS with 16.20 ms latency versus DiskANN's 2,597 QPS and 49.06 ms: a 3× latency reduction driven by fewer redundant disk accesses. Figure 2b confirms that MCGI matches DiskANN's recall trajectory across all tested beam widths L, while Figure 2c shows reduced tail latency at high-recall operating points.

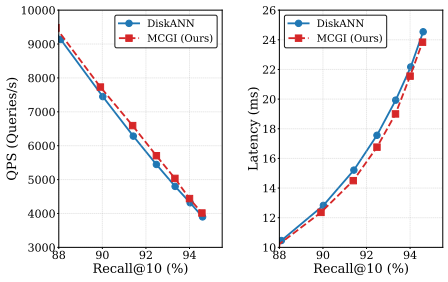
Table 1: Peak QPS at $\geq$ 95% recall. †memory-resident, as upper bounds.

Algorithm	Locality	QPS
PipeANN-R32 [11]	disk	52.2
PipeANN-R48 [11]	disk	60.3
DiskANN [36]	disk	64.7
MCGI (this work)	disk	375.1
IVF Flat† [20]	memory	590.5
CSPG-R32† [41]	memory	738.2
CSPG-R48† [41]	memory	1,029
HNSW† [29]	memory	2,165

We further evaluated MCGI on the T2I-1B dataset in Figure 3. The performance gap narrows compared to SIFT1B as expected: the I/O bottleneck is less pronounced when the compute cost per hop dominates. Nevertheless, MCGI maintains a consistent advantage at high-recall operating points (Recall@10 > 90%), confirming that manifold-aware pruning reduces redundant distance computations in CPU-bound scenarios.

4.4 Comparison with End-to-end Indexing Systems

To position MCGI within the landscape of state-of-the-art graph indexing systems, we conduct a head-to-head comparison with CSPG [41] and PipeANN [11, 12]. As illustrated in Figure 4, we evaluate these systems under identical storage budgets by fixing the maximum out-degree at $R = 32$ and $R = 48$. The results reveal a fundamental performance gap between MCGI and existing baselines on the high-dimensional GIST1M manifold. Specifically, while PipeANN employs aggressive I/O pipeline parallelism via io_uring to mask SSD latency, its search performance suffers a catastrophic



(a) Throughput (High Recall) (b) Latency (High Recall)

Figure 3: Trade-off on T2I-1B.

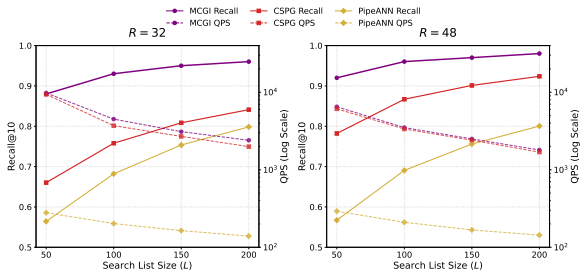


Figure 4: Performance comparison between MCGI, CSPG, and PipeANN on GIST1M under degree constraints (L) and two Rs.

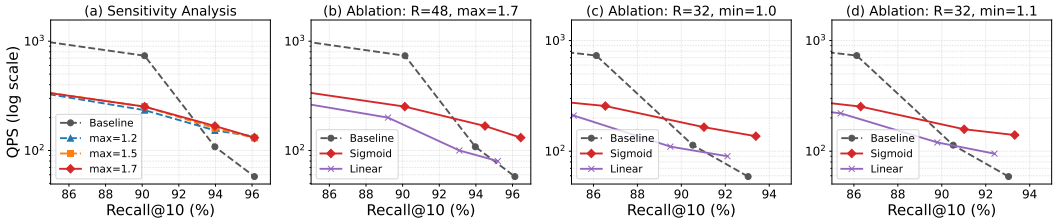


Figure 5: Ablation and sensitivity analysis of the mapping function and pruning parameters.

collapse due to a fragmented topology. Under the $R = 32$ configuration, PipeANN struggles to surpass an 80% recall ceiling even with an enlarged search pool ($L = 200$), incurring excessive I/O overhead (averaging > 220 IOs per query). In contrast, MCGI achieves a near-perfect recall of 96.02% under the same constraint, outperforming PipeANN in both accuracy and throughput.

4.5 Ablation Study and Range Sensitivity

We evaluate the range sensitivity of MCGI to the dynamic pruning upper bound, α_{max}, which governs the trade-off between graph connectivity and distance computation overhead. As depicted in Figure 5(a), tuning α_{max} from 1.2 to 1.7 provides a smooth and predictable Pareto frontier for the QPS-Recall tradeoff. While a lower bound ($\alpha_{max} = 1.2$) limits connectivity, increasing the bound allows MCGI to preserve critical manifold paths that are otherwise prematurely pruned. Specifically, at $\alpha_{max} = 1.7$, MCGI successfully breaks the baseline's recall ceiling (reaching 96.45%) while maintaining more than double the search throughput (>130 QPS) of the baseline.

To further validate the necessity of our non-linear density mapping Sigmoid, we conduct an ablation study comparing our proposed LID-aware Sigmoid transformation against a vanilla Linear mapping. As shown in Figures 5(b)-(d) across three distinct topological configurations, replacing the Sigmoid function with a naive Linear scaling results in a severe performance collapse in the extreme high-recall setup. For instance, under the $R = 48$ configuration, the Linear mapping suffers a nearly 50% drop in QPS and a noticeable degradation in recall compared to Sigmoid.

5 Conclusion

This paper presents MCGI, a geometry-aware disk-resident indexing method for high-dimensional data of varying intrinsic dimensionality. MCGI resolves the Euclidean-Geodesic mismatch by adapting graph topology to local intrinsic dimensionality. Unlike existing methods that treat all regions of the underlying data manifold uniformly, MCGI dynamically allocates graph connectivity based on local geometric complexity, achieving what static-parameter approaches cannot: manifold-consistent routing that remains efficient as dimensionality scales and fluctuates. Our theoretical analysis formalizes this advantage with provable guarantees, and our empirical results confirm its effectiveness in practice: $5.8\times$ higher throughput at 95% recall on high-dimensional datasets (e.g., 960-dimensional GIST1M) and $3\times$ lower latency on billion-scale datasets (e.g., 1-billion SIFT1B), narrowing the long-standing performance gap between disk-resident and in-memory indices.

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A Technical appendices and supplementary material

A.1 Complete Proofs

Proposition A.1 (Monotonicity)**.** *The mapping function Φ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{15}$$

Proof. Let $L = \widehat{\mathrm{LID}}(u)$ be the independent variable. We define the normalized Z-score z as a function of L:

$$z(L) = \frac{L - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{16}$$

The mapping function is defined as:

$$\Phi(L) = \alpha_{\min} + \frac{C}{1 + \exp(z(L))}, \tag{17}$$

where $C = \alpha_{\max} - \alpha_{\min}$. Since we strictly require $\alpha_{\max} > \alpha_{\min}$, it follows that $C > 0$. To determine the sign of the gradient, we apply the chain rule:

$$\frac{d\Phi}{dL} = \frac{d\Phi}{dz} \cdot \frac{dz}{dL}. \tag{18}$$

First, we differentiate the Z-score term with respect to L:

$$\frac{dz}{dL} = \frac{1}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{19}$$

Next, we differentiate the logistic component Φ with respect to z:

$$\frac{d\Phi}{dz} = \frac{d}{dz}\left(\alpha_{\min} + C(1 + e^z)^{-1}\right) \tag{20}$$

$$= C \cdot (-1) \cdot (1 + e^z)^{-2} \cdot \frac{d}{dz}(1 + e^z) \tag{21}$$

$$= -C \cdot \frac{e^z}{(1 + e^z)^2}. \tag{22}$$

Combining these terms yields the full derivative:

$$\frac{d\Phi}{dL} = -\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1 + e^z)^2}. \tag{23}$$

We analyze the sign of each component:

- The operational range constant $C > 0$.
- The standard deviation $\sigma_{\widehat{\mathrm{LID}}} > 0$, assuming the dataset exhibits non-zero geometric variance.
- The exponential function $e^z > 0$ for all $z \in \mathbb{R}$.
- The denominator $(1 + e^z)^2 > 0$.

Therefore, the term $\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \frac{e^z}{(1+e^z)^2}$ is strictly positive. The leading negative sign guarantees that $\frac{d\Phi}{dL} < 0$. This confirms that the pruning parameter α strictly decreases as the local geometric complexity increases, thereby enforcing a more conservative graph topology in high-LID regions to prevent topology distortion in complex regions. □

Proposition A.2 (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. For any node u with a finite LID estimate:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{24}$$

Proof. Let $S(u)$ denote the logistic component of the mapping function:

$$S(u) = \frac{1}{1 + \exp(z(u))}. \tag{25}$$

For any finite input $\widehat{\text{LID}}(u)$, the Z-score $z(u)$ is finite. The exponential function maps the real line to the positive real line, i.e., $\exp(z(u)) \in (0, \infty)$. Consequently, the denominator lies in the interval $(1, \infty)$. Taking the reciprocal yields the bounds for the logistic component:

$$0 < S(u) < 1. \tag{26}$$

Substituting $S(u)$ back into the definition of Φ:

$$\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u). \tag{27}$$

Since $(\alpha_{\max} - \alpha_{\min}) > 0$, we can apply the inequality boundaries:

$$\alpha(u) > \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 0 = \alpha_{\min}, \tag{28}$$

$$\alpha(u) < \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 1 = \alpha_{\max}. \tag{29}$$

This proves that the topology is strictly confined. The pruning behavior never exceeds the relaxation upper limit ($\alpha_{\max}$) and never becomes stricter than the lower limit ($\alpha_{\min}$), ensuring graph connectivity and preventing degree explosion. □

Lemma A.3 (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query q on a manifold $\mathcal{M}$, the expected number of distance evaluations N_{dist} required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = \textit{LID}(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{30}$$

where $\lambda > 0$ is a geometric constant derived from the required convergence rate.

Proof. Let the local dataset around a query q be modeled as a uniform distribution on a manifold of intrinsic dimension d. Consider a greedy routing step where the algorithm is currently at a node u with distance $r = \|u - q\|$.

1. Geometric Condition for Improvement. To make progress, the algorithm must identify a neighbor v such that the distance to the query is reduced by a factor ϵ (where $0 < \epsilon < 1$), i.e., $\|v - q\| \leq (1 - \epsilon)r$. Let $s = \|u - v\|$ be the distance between the current node and the candidate neighbor (the step size). Applying the Law of Cosines to triangle $\triangle uvq$, we have:

$$\|v - q\|^2 = r^2 + s^2 - 2rs\cos\theta, \tag{31}$$

where θ is the angle between the vector $\vec{uq}$ and $\vec{uv}$. The improvement condition $\|v-q\|^2 \leq (1-\epsilon)^2 r^2$ implies:

$$\begin{aligned} r^2 + s^2 - 2rs\cos\theta &\leq (1-\epsilon)^2 r^2, \\ 2rs\cos\theta &\geq r^2 + s^2 - r^2(1-\epsilon)^2, \\ \cos\theta &\geq \frac{r^2 + s^2 - r^2(1-\epsilon)^2}{2rs}. \end{aligned} \tag{32}$$

Assuming the step size is small relative to the distance (i.e., $s \ll r$), the right-hand side is dominated by $1 - (1-\epsilon)^2 \approx 2\epsilon$, yielding a critical angle threshold θ_{max} such that any successful neighbor must fall within the cone defined by $0 \leq \theta \leq \theta_{max}$.

2. Volume Concentration of the Improving Region. In the tangent space at u, the directions of neighbors are uniformly distributed on the unit hypersphere $\mathbb{S}^{d-1}$. The probability $P_{success}$ of finding a neighbor in the improving cone corresponds to the ratio of the surface area of the spherical cap $\mathcal{C}(\theta_{max})$ to the total surface area of $\mathbb{S}^{d-1}$. This ratio is given by the regularized incomplete beta function, which can be expressed in terms of the integral over the polar angle ϕ:

$$P_{success} = \frac{\text{Area}(\mathcal{C}(\theta_{max}))}{\text{Area}(\mathbb{S}^{d-1})} = \frac{\int_0^{\theta_{max}} \sin^{d-2}\phi \, d\phi}{\int_0^{\pi} \sin^{d-2}\phi \, d\phi}. \tag{33}$$

3. Asymptotic Analysis. We analyze the asymptotic behavior of this ratio as $d \to \infty$. The denominator relates to the surface area of the hypersphere and can be evaluated using the standard integral identity for powers of sine. Exploiting the symmetry of $\sin\phi$ around $\pi/2$, we have:

$$\begin{aligned}\int_0^{\pi} \sin^{d-2}\phi \, d\phi &= 2\int_0^{\frac{\pi}{2}} \sin^{d-2}\phi \, d\phi \\ &= 2 \cdot \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})} = \sqrt{\pi}\frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})}.\end{aligned} \tag{34}$$

For high-dimensional spaces ($d \gg 1$), we apply the asymptotic property of the Gamma function ratio, $\frac{\Gamma(x+a)}{\Gamma(x)} \approx x^a$. Setting $x = d/2$ and $a = -1/2$, we obtain the approximation:

$$\sqrt{\pi}\frac{\Gamma(\frac{d}{2} - \frac{1}{2})}{\Gamma(\frac{d}{2})} \approx \sqrt{\pi}\left(\frac{d}{2}\right)^{-\frac{1}{2}} = \sqrt{\frac{2\pi}{d}}. \tag{35}$$

For the numerator, we exploit the concentration of measure near the integration boundary θ_{max}. Since the integrand $\sin^{d-2}\phi$ decays exponentially fast away from θ_{max}, the integral is dominated by the contribution near the upper limit. We apply integration by parts to extract the leading order term. Recall that $\frac{d}{d\phi}(\sin^{d-1}\phi) = (d-1)\sin^{d-2}\phi\cos\phi$. We rewrite the integrand as:

$$\int_0^{\theta_{max}} \sin^{d-2}\phi \, d\phi = \int_0^{\theta_{max}} \frac{1}{(d-1)\cos\phi}\frac{d}{d\phi}(\sin^{d-1}\phi)\, d\phi. \tag{36}$$

Neglecting lower-order terms generated by the derivative of $\frac{1}{\cos\phi}$, the integral approximates to the boundary term:

$$\left[\frac{\sin^{d-1}\phi}{(d-1)\cos\phi}\right]_0^{\theta_{max}} = \frac{\sin^{d-1}\theta_{max}}{(d-1)\cos\theta_{max}}. \tag{37}$$

Substituting these approximations back into the probability ratio, we obtain:

$$P_{success} \approx \frac{1}{\sqrt{2\pi}\cos\theta_{max}} \cdot \frac{\sqrt{d}}{d-1} \cdot (\sin\theta_{max})^{d-1}. \tag{38}$$

For large d, the term $\frac{\sqrt{d}}{d-1}$ is asymptotically equivalent to $d^{-1/2}$. Since θ_{max} is fixed by the geometric constraints, the trigonometric factors are constant relative to d. Let $\gamma = \sin\theta_{max} < 1$. The probability scales as:

$$P_{success} \propto d^{-1/2} \cdot \gamma^d = d^{-1/2}\exp(d\ln\gamma). \tag{39}$$

Defining the decay rate $\lambda = -\ln\gamma > 0$ (since $\gamma < 1$), we strictly obtain:

$$P_{success} \propto d^{-1/2}\exp(-\lambda \cdot d). \tag{40}$$

4. Complexity Bound. The expected number of distance evaluations follows $\mathbb{E}[N_{dist}] = 1/P_{success}$. Thus:

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\sqrt{d} \cdot \exp(\lambda \cdot d)\right). \tag{41}$$

This confirms that the routing cost is dominated by the exponential term $\exp(\lambda \cdot \mathrm{LID}(q))$, necessitating the adaptive beam width design in MCGI. □

Proposition A.4 (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability δ across the manifold (i.e., $\mathbb{P}(\text{fail} \mid q) \leq \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda \cdot \mathit{LID}(q)\right). \tag{42}$$

Proof. Consider a beam search with width L. The probability of failing to find a descent step in one round is approximately $(1 - P_{success})^L$. To guarantee a recall target, we enforce a constant upper bound δ on the failure probability:

$$(1 - P_{success})^L \leq \delta. \tag{43}$$

Taking the natural logarithm and using the approximation $\ln(1-x) \approx -x$ for small $P_{success}$, we require:

$$-L \cdot P_{success} \leq \ln \delta \implies L \geq \frac{-\ln \delta}{P_{success}}. \tag{44}$$

Substituting the relationship $P_{success} = 1/\mathbb{E}[N_{dist}]$ established in Lemma 3.5, we obtain the asymptotic upper bound $P_{success} \leq \mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))$, where $d = \text{LID}(q)$. To satisfy the recall constraint in the worst case scenario, the search budget L must grow inversely to $P_{success}$:

$$L \geq \frac{-\ln \delta}{\mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))} = \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right). \tag{45}$$

In high-dimensional spaces ($d \gg 1$), the exponential growth term strictly dominates the polynomial decay factor $1/\sqrt{d}$. By absorbing the lower-order polynomial variations and the confidence factor $-\ln \delta$ into a generalized constant C_δ, the necessary budget asymptotically simplifies to:

$$L(q) \geq C_\delta \cdot \exp(\lambda \cdot \text{LID}(q)). \tag{46}$$

□

Proposition A.5 (Connectivity Preservation)**.** *Let G_{EMST} and G_{RNG} denote the Euclidean Minimum Spanning Tree and the Relative Neighborhood Graph, respectively. For any configuration of points in general position, provided that $\alpha(u) \geq 1.0$ for all $u \in V$, the graph G_{MCGI} satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{47}$$

Consequently, G_{MCGI} is connected.

Proof. The proof proceeds in two steps. First, we invoke the classical result established by Toussaint [38], which proves that the Relative Neighborhood Graph (RNG) is a supergraph of the Euclidean Minimum Spanning Tree (EMST) for any finite set of points in a metric space:

$$E_{EMST} \subseteq E_{RNG}. \tag{48}$$

Since the EMST connects all vertices in a single component, G_{RNG} is necessarily connected.

Second, we show that $E_{RNG} \subseteq E_{MCGI}$. In the RNG, an edge between u and v is pruned if and only if a witness node falls within the intersection of two spheres centered at u and v with radius $d(u, v)$. We denote this geometric intersection as the exclusion region $\mathcal{R}_{RNG}$. Therefore, an edge exists in the RNG if and only if $\mathcal{R}_{RNG}$ contains no data points. In MCGI, the pruning condition requires the witness node to be within a distance of $\frac{1}{\alpha(u)} d(u, v)$ from v. Because the algorithm enforces $\alpha(u) \geq 1.0$, the pruning radius in MCGI is strictly less than or equal to the pruning radius of the RNG. Consequently, the exclusion region of MCGI is a geometric subset of the RNG exclusion region:

$$\mathcal{R}_{MCGI} \subseteq \mathcal{R}_{RNG}. \tag{49}$$

This subset relationship of exclusion regions creates an inverse superset relationship for the preserved edges. If an edge survives the RNG condition, its $\mathcal{R}_{RNG}$ is empty. Since $\mathcal{R}_{MCGI}$ is contained entirely within $\mathcal{R}_{RNG}$, it must also be empty. Thus, any edge present in the RNG is guaranteed to be preserved in MCGI, yielding the inclusion hierarchy of edge sets:

$$E_{RNG} \subseteq E_{MCGI}. \tag{50}$$

By combining this with the classical result that $E_{EMST} \subseteq E_{RNG}$ derived in [38], we establish the complete global connectivity chain. □

Proposition A.6 (Estimation Robustness)**.** *Let $\widehat{\text{LID}}(u)$ be the MLE estimate computed with k nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \text{LID}(u)}{4\, \sigma_{\widehat{\text{LID}}} \sqrt{k}}. \tag{51}$$

Proof. Let $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ denote the estimation error. By the first-order Taylor expansion of Φ around $\mathrm{LID}(u)$:

$$|\Delta\alpha(u)| = \left|\Phi\Big(\widehat{\mathrm{LID}}(u)\Big) - \Phi(\mathrm{LID}(u))\right| \leq \sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|. \tag{52}$$

From the derivative computed in Proof A.1, the magnitude of the gradient is:

$$\left|\frac{d\Phi}{dL}\right| = \frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1+e^z)^2}, \tag{53}$$

where $C = \alpha_{\max} - \alpha_{\min}$. The logistic factor $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $1/4$ at $z = 0$, so:

$$\left|\frac{d\Phi}{dL}\right| \leq \frac{\alpha_{\max} - \alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}. \tag{54}$$

Substituting the asymptotic error bound from Eq. (11), namely $|\epsilon| = O(\mathrm{LID}(u)/\sqrt{k})$ [26], yields:

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{55}$$

This completes the proof. □

Proposition A.7 (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies Eq.* (14)*, which grows exponentially with the deviation of* $\mathrm{LID}(q)$ *from the dataset mean.*

Proof. Consider a beam search with fixed width $\bar{L}$. From Proof A.4, the failure probability at query q is:

$$P(\text{fail} \mid q) = \left(1 - P_{\text{success}}(q)\right)^{\bar{L}}. \tag{56}$$

From Lemma 3.5, we have:

$$P_{\text{success}}(q) \propto d^{-1/2} \exp(-\lambda \cdot d), \quad d = \mathrm{LID}(q). \tag{57}$$

For the oracle schedule $L^*(q)$ in Definition 3.9, the constraint $P(\text{fail} \mid q) \leq \delta$ holds by construction. For fixed $\bar{L}$, define $\Delta L(q) = L^*(q) - \bar{L}$. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\text{avg}}$, we have $\Delta L(q) > 0$ and:

$$\begin{aligned} P(\text{fail} \mid q) - \delta &= (1 - P_{\text{success}})^{\bar{L}} - (1 - P_{\text{success}})^{L^*(q)} \\ &\geq 1 - \exp\left(-\frac{C_\delta}{\bar{L}} \exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\text{avg}}\big)\big)\right), \end{aligned} \tag{58}$$

where the inequality applies $1 - e^{-x} \leq x$ and substitutes $L^*(q)$ from Definition 3.9. The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, confirming the exponential widening of the recall gap. □

Corollary A.8 (Construction-time Structural Compensation)**.** *Under any fixed beam width $\bar{L}$, the adaptive pruning in MCGI monotonically reduces $P(\text{fail} \mid q)$ for queries in high-LID regions compared to any static-α baseline operating under the same $\bar{L}$.*

Proof. We proceed in two steps: first showing that MCGI preserves a strictly larger edge set in high-LID regions, then connecting this to a lower failure probability under fixed beam search.

Step 1: Edge set inclusion. The pruning criterion in MCGI discards an edge (u, v) if and only if there exists a witness node n such that $\alpha(u) \cdot d(n, v) \leq d(u, v)$. For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\text{avg}}$, Proposition A.1 guarantees $\alpha(u) < \bar{\alpha}$. Since $\alpha(u) < \bar{\alpha}$ strictly shrinks the left-hand side of the pruning condition, any edge that would be pruned under $\alpha(u)$ is also pruned under $\bar{\alpha}$, but not vice versa. Therefore $\mathcal{E}_{\text{MCGI}} \supseteq \mathcal{E}_{\bar{\alpha}}$ in high-LID regions, with strict inclusion whenever at least one edge survives $\alpha(u)$ but not $\bar{\alpha}$.

Step 2: Failure probability. Under beam search with fixed width $\bar{L}$, the failure probability at query q is governed by $P_{\text{success}}(q)$, the probability of finding a descent step from any node on the routing path (Proof A.4). A strictly larger outgoing edge set at each high-LID node monotonically increases

the number of candidate directions evaluated per beam step, which by the same geometric argument in Proof A.4 yields a weakly higher $P_{\text{success}}(q)$ and thus:

$$P(\text{fail} \mid q, \text{MCGI}) \leq P(\text{fail} \mid q, \bar{\alpha}), \quad \forall\, q \text{ s.t. } \text{LID}(q) > \text{LID}_{\text{avg}}, \tag{59}$$

with equality only when $\alpha(u) = \bar{\alpha}$ for all u on the routing path, i.e., when the data is geometrically uniform. This establishes a monotone improvement over fixed-α baselines; quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) is left for future work. □

A.2 Additional Experimental Details

Experiment Platform All experiments were conducted on the Chameleon Cloud [25] platform using a compute node equipped with 96 AMD EPYC 7352 CPU cores and 256 GiB of RAM. The operating system is Ubuntu 24.04 LTS. All algorithms were compiled using GCC 11.4 with -O3 and AVX2 optimizations enabled to fully utilize the hardware instruction set. We utilize a single enterprise-grade Samsung PM1733 NVMe SSD (model MZWLJ7T6HALA) with a total physical capacity of 7.68 TB. In this configuration, the physical drive is divided into four independent hardware namespaces, each providing approximately 1.7 TiB of storage capacity. These four namespaces are subsequently aggregated into a single 7 TB logical volume using the Linux Logical Volume Manager (LVM). This setup leverages the PCIe Gen4 interface to deliver the massive I/O throughput and low latency.

Datasets We evaluate our method on five standard benchmarks, ranging from million-scale to billion-scale, to comprehensively test robustness across varying intrinsic dimensionalities and data volumes. In the following descriptions, N denotes the number of base vectors and D represents the vector dimensionality:

- SIFT1M [21] ($N = 10^6, D = 128$): A standard computer vision dataset using Euclidean distance (L_2).
- GloVe-100 [30] ($N = 1.2 \times 10^6, D = 100$): Word embedding vectors measuring semantic similarity. Following standard practice, we normalize the vectors to unit length and use Euclidean distance as a proxy for Cosine similarity.
- GIST1M [21] ($N = 10^6, D = 960$): A high-dimensional dataset representing global image features. This dataset is particularly challenging for index structures due to the sparsity of the space and the high intrinsic dimensionality.
- SIFT1B [22] ($N = 10^9, D = 128$): A billion-scale dataset used to evaluate the scalability and I/O efficiency of our method. It shares the same feature distribution as SIFT1M but scales the volume by three orders of magnitude, representing a realistic industrial scenario.
- T2I-1B [34] ($N = 10^9, D = 200$): A billion-scale dataset consisting of CLIP [31] embeddings derived from the LAION-5B dataset. Unlike visual descriptors like SIFT, T2I-1B represents a cross-modal retrieval scenario with a more complex manifold structure and higher intrinsic dimensionality, posing greater challenges for graph-based indexing algorithms.

Baselines We compare MCGI against five baselines, each serving a distinct role:

- PipeANN [11, 12]: A recent state-of-the-art disk-based system that optimizes search throughput via aggressive I/O pipeline parallelism. We include it to evaluate whether system-level asynchronous I/O optimizations can compensate for topological deficiencies in high-dimensional spaces.
- CSPG [41]: A high-performance C++ implementation of proximity graphs designed for efficiency. It serves as a rigorous algorithmic baseline to benchmark our LID-aware pruning strategy against standard graph construction techniques.
- DiskANN (Vamana) [36]: The standard disk-based graph index in Faiss. Since MCGI builds upon the Vamana architecture, this comparison isolates the specific gains derived from our manifold-aware routing strategy.
- IVF-Flat (Faiss) [20]: An in-memory inverted index serving as a performance roofline. Although running in memory-mapped mode, its sequential scanning pattern allows aggressive

OS caching, effectively simulating in-memory performance. We include it to benchmark how closely our disk-resident solution approaches the throughput limits of memory-resident systems.

- HNSW [29]: The state-of-the-art in-memory graph index known for its superior recall-latency trade-off. Unlike disk-resident approaches, HNSW maintains the entire hierarchical graph topology in DRAM, completely eliminating I/O latency. We include it to establish a performance ceiling for graph-based search, allowing us to quantify the gap between our disk-resident solution and the theoretical limit of unconstrained memory-resident traversal.

Evaluation Metrics We adhere to the standard evaluation protocol for ANN search. We measure Recall@10 against QPS (Queries Per Second). Additionally, we report the query latency at critical high-recall operating points (e.g., 95% Recall) to assess the tail latency characteristics.

Build Configuration Table 2 summarizes the dataset specifications and indexing parameters used during the construction of MCGI and baseline DiskANN indices. R denotes the maximum outdegree of each node in the graph. L_{build} is the size of the candidate list during index construction. α Range indicates the range of pruning aggressiveness explored during the LID-based Sigmoid mapping. m_{PQ} specifies the number of bytes allocated for Product Quantization (PQ) codes when applicable. Data Type indicates whether the dataset vectors are stored as 32-bit floating-point numbers or 8-bit unsigned integers.

Table 2: Dataset Specifications and Indexing Parameters.

Dataset	R	L_{build}	α Range	m_{PQ} (Byte)	Data Type
SIFT1M	{32, 48, 64}	100	$[1.0, 1.5]$	N/A	float32
GloVe-100	{32, 48, 64}	100	$[1.0, 1.5]$	N/A	float32
GIST1M	{32, 48, 64, 96}	150	$[1.0, 1.5]$	N/A	float32
SIFT1B	{32, 48}	50	$[1.0, 1.5]$	16	uint8
T2I-1B	{32, 48}	50	$[1.0, 1.5]$	16	float32

Search Configuration Table 3 lists the statistical parameters derived from the dataset's LID distribution used in the Sigmoid mapping function $\Phi(\cdot)$, alongside the runtime search sweep parameters. μ_{LID} and σ_{LID} denote the mean and standard deviation of the Local Intrinsic Dimensionality (LID) estimates across the dataset. Beam (L_{search}) indicates the range of candidate list sizes evaluated during query execution to capture the Recall-QPS trade-off. Threads specifies the number of parallel threads utilized during search to maximize hardware concurrency.

Table 3: Runtime Search Parameters and LID Statistics.

Dataset	μ_{LID} (Mean)	σ_{LID} (Std)	Beam (L_{search})	Threads
SIFT1M	14.2	3.1	$10 \rightarrow 200$	96
GloVe-100	18.5	4.2	$10 \rightarrow 200$	96
GIST1M	22.1	5.8	$10 \rightarrow 200$	96
SIFT1B	19.5	7.9	$10 \rightarrow 200$	96
T2I-1B	18.3	7.0	$10 \rightarrow 200$	96

A.3 Algorithm Pseudo-Code

Algorithm 1 outlines the offline construction process of the indexing structure. The procedure is divided into two sequential phases.

- The first phase focuses on geometric calibration by calculating the Local Intrinsic Dimensionality for all vectors in the dataset in parallel. After obtaining the global mean and standard deviation of these estimates, the algorithm assigns a personalized pruning parameter α_u to each node. This parameter is derived using a logistic function that maps the standardized geometric complexity of the local neighborhood into a bounded operational range.

- The second phase handles topology refinement. Starting from a random graph, the algorithm iteratively optimizes the connections. For each node, it executes a greedy search to identify a pool of candidate neighbors. It then evaluates these candidates alongside existing neighbors using a dynamic occlusion criterion. A candidate is pruned if a previously accepted neighbor falls within a restricted exclusion region defined by the precomputed α_u. Survivors are added to the updated neighbor list until the degree limit is reached.

Algorithm 1 Manifold-Consistent Graph Indexing (MCGI)

Input: Dataset X, Max Degree R, Beam Width L
Output: Optimized Graph G

// Phase 1: Geometric Calibration
$\mathcal{L} \leftarrow$ ParallelEstimateLID(X)
$\mu \leftarrow$ Mean($\mathcal{L}$)
$\sigma \leftarrow$ StdDev($\mathcal{L}$)
for each node $u \in V$ in parallel **do**
  $z_u \leftarrow (\mathcal{L}[u] - \mu)/\sigma$
  $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
end for

// Phase 2: Topology Refinement
$G \leftarrow$ RandomGraph(X, R)
for $iter \leftarrow 1$ to $MaxIter$ **do**
  for each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch(u, G, L)
    $\mathcal{N}_{new} \leftarrow \emptyset$
    for $v \in$ SortByDistance($\mathcal{C} \cup \mathcal{N}(u)$) **do**
      $pruned \leftarrow$ False
      for $n \in \mathcal{N}_{new}$ **do**
        if $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        end if
      end for
      if not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add(v)
      end if
    end for
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  end for
end for

Algorithm 2 presents an optimized online variant designed to handle massive datasets where exhaustive geometric profiling is computationally prohibitive.

- It begins with a fast approximation phase that samples a small subset of the data. By evaluating this subset, the algorithm quickly bootstraps the global mean and standard deviation of the underlying probability distribution. This step provides a coarse but efficient estimate of the overall geometric complexity, enabling the algorithm to initialize the pruning parameters without incurring the cost of a full dataset pass.
- The second phase merges the geometric estimation directly into the iterative topology refinement loop. Instead of computing all complexities beforehand, the algorithm dynamically estimates the node specific dimensionality on the fly using the candidate set discovered during the greedy search step. This real time estimate is immediately used to calculate the specific pruning parameter α_u. The algorithm then applies this dynamic parameter to the pruning criterion, determining which edges to preserve and which to discard. This approach ensures the graph connectivity adapts to the manifold geometry progressively as the quality of the candidate neighbors improves with each iteration.

Algorithm 2 Online Manifold-Consistent Graph Indexing

Input: Dataset X, Max Degree R, Beam Width L, Sample Rate S
Output: Optimized Graph G

// Phase 1: Bootstrap Statistics (Fast Approximation)
$X_{sample} \leftarrow$ RandomSample(X, S)
$\mathcal{L}_{sample} \leftarrow$ EstimateLID(X_{sample}) {Compute LID on subset}
$\mu \leftarrow$ Mean$(\mathcal{L}_{sample})$, $\sigma \leftarrow$ StdDev$(\mathcal{L}_{sample})$

// Phase 2: Online Topology Refinement
$G \leftarrow$ RandomGraph(X, R)
for $iter \leftarrow 1$ to $MaxIter$ **do**
  for each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch(u, G, L)
    // Online LID Estimation & Parameter Adaptation
    $\hat{\ell}_u \leftarrow$ ComputeLID$(u, \mathcal{C})$ {Estimate from neighbors}
    $z_u \leftarrow (\hat{\ell}_u - \mu)/\sigma$
    $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
    $\mathcal{N}_{new} \leftarrow \emptyset$
    for $v \in$ SortByDistance$(\mathcal{C} \cup \mathcal{N}(u))$ **do**
      $pruned \leftarrow$ False
      for $n \in \mathcal{N}_{new}$ **do**
        // Pruning with dynamic α_u
        if $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        end if
      end for
      if not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add(v)
      end if
    end for
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  end for
end for